Experiment-6 Parseval's Relation and Harmonic Spectrum Analysis

OBJECTIVE:

(i)To verify Parseval’s relation for continuous-time periodic signals and analyze the relationship between time-domain and frequency-domain power.

(iv) To compare the total power obtained in the time domain with the harmonic power distribution in the frequency domain.

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Software / Laboratory Environment:

MATLAB

MATLAB Coding:

```
% Plot comparison of power values

figure('Color', 'w');

power_values = [P_time, P_freq];

b = bar(power_values, 0.55, ...
    'FaceColor', 'flat');

% Different colors for both bars
b.CData(1,:) = [0.2 0.5 0.8];
b.CData(2,:) = [0.9 0.4 0.2];

% X-axis labels
set(gca, 'XTick', [1 2]);
set(gca, 'XTickLabel', ...
    {'Time Domain', 'Frequency Domain'});
```

```
% Add numerical values above bars
text(1, P_time, sprintf('%.6f', P_time), ...
    'HorizontalAlignment', 'center', ...
    'VerticalAlignment', 'bottom');

text(2, P_freq, sprintf('%.6f', P_freq), ...
    'HorizontalAlignment', 'center', ...
    'VerticalAlignment', 'bottom');

% Graph formatting
ylabel('Average Power');
xlabel('Domain');

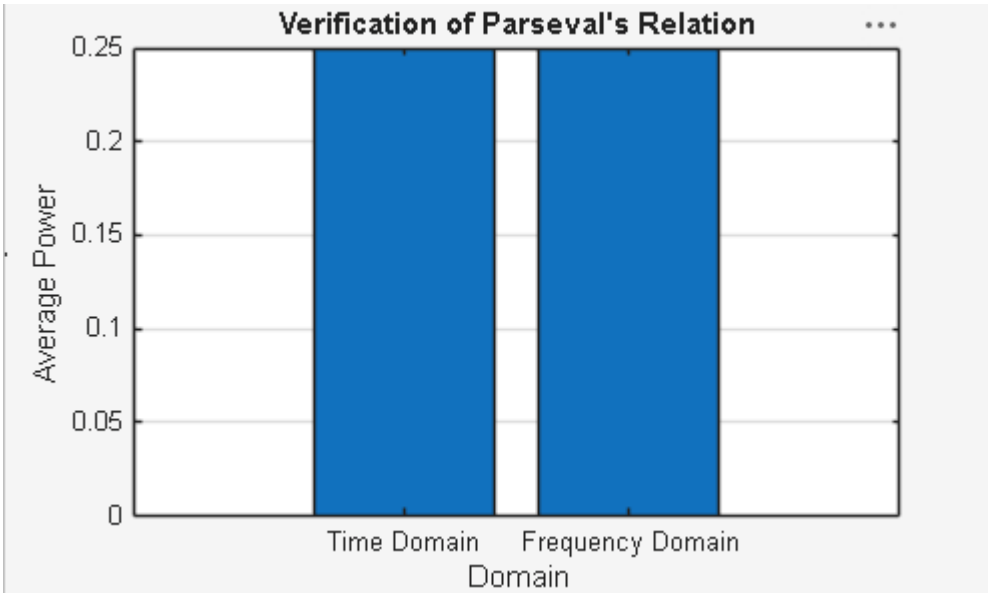
title('Verification of Parseval''s Relation');

ylim([0 max(power_values)*1.15]);

grid on;
box on;

set(gca, 'FontSize', 12);
```

OUTPUT:



RESULT:

The experiment successfully verified Parseval's relation for continuous-time periodic signals by demonstrating the equivalence of power calculated in the time and frequency domains. The distribution of power among the fundamental and harmonic components was analyzed through the harmonic spectrum, and the total time-domain power was found to be consistent with the sum of harmonic powers in the frequency domain.

RUBRICS:

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightage**	**Marks(10)**	**Marks obtained**

(i) To verify Parseval's relation for continuous-time periodic signals and analyze the relationship between time-domain and frequency-domain power.	**Fourier Series Coefficient Computation**	Correctly identifies the fundamental period T0 and frequency ω0 of the periodic signal. Computes the Fourier series coefficients (c_k, or a_k and b_k) accurately, analytically or in MATLAB, and plots the signal with its harmonic reconstruction with correct labels and scaling.	Computes the coefficients correctly with minor errors in the period, limits, scaling or labelling.	Computes only some coefficients correctly, or has noticeable errors in T0, integration or code, and needs guidance.	Unable to determine the period or compute the Fourier series coefficients.	**20**	2	
	Time-Domain Power & Parseval	Correctly computes the average power P .Applies Parseval's relation	Computes the power and applies Parseval's relation	Computes the time-domain power or applies Parseval'	Unable to compute the power or verify Parseval	**20**	2	

	Verification	correctly, verifies that the two values agree within a small numerical error, and explains the time-frequency relationship clearly.	correctly, with minor numerical or interpretation errors.	s relation only partly. Verification is incomplete or has noticeable errors, and needs guidance.	's relation.			
(ii) To compare the total power obtained in the time domain with the harmonic power distribution in the frequency domain.	**Harmonic Power Distribution**	Correctly computes the power in each harmonic and the DC power. Plots the power spectrum (stem plot) with labelled axes, shows the cumulative power, and identifies the dominant harmonics accurately.	Computes the harmonic powers and plots the spectrum correctly, with minor errors in indexing, scaling or labelling.	Computes some harmonic powers but has noticeable errors in the spectrum or the DC and harmonic terms, and needs assistance.	Unable to compute or plot the harmonic power distribution.	**20**	2	

	Time vs Frequency Power Comparison	Compares total time-domain power with the sum of harmonic powers for an increasing number of harmonics. Reports the percentage error and the power captured, explains convergence and the effect of truncation (for example for square or triangular waves), and draws correct conclusions.	Compares the total powers correctly, with minor errors in the error calculation or limited explanation of convergence.	Makes a basic comparison with incomplete error analysis, or gives limited explanation, and needs guidance.	Unable to compare the time-domain and frequency-domain power correctly.	**20**	2	

Interpret the results for both objectives and maintain a proper technical record.	**Engineering Analysis, Record & Interpretation**	Provides technically accurate interpretation of the results, relating power to harmonic content and signal waveform. Maintains a complete record with aim, theory, MATLAB code, plots, observations, result and conclusion, with clear organization.	Maintains a proper record and gives mostly accurate interpretation, with minor omissions in explanation or presentation.	Record and interpretation are basic, with partial explanation or missing elements.	Incomplete record, limited or incorrect interpretation of the results.	**20**	2	
					Total	**100**	**10**	